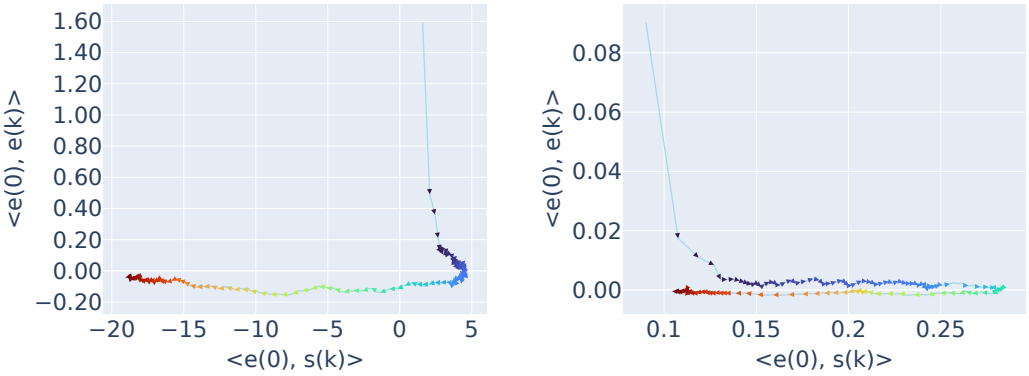


(a) PI (b) PID

Figure 6: Scalar errors across time step of randomly initialized model after applying PI and PID controller. Colors from blue to red denote the time (layer) dimension.

**Proposition 4 (Agressive PI gain leads to a large first overshoot)** *Let $k_0$ be the first sign-change time, i.e.,*

$$\boldsymbol{e_v}(j) \geq 0 \ \forall j = 0,1,\ldots,k_0-1, \qquad \boldsymbol{e_v}(k_0) < 0,$$

*and let $k_1$ be the first time the trajectory returns to the nonnegative side,*

$$\boldsymbol{e_v}(k_1) \geq 0, \qquad \boldsymbol{e_v}(i) < 0 \ \forall i = k_0, k_0+1, \ldots, k_1-1.$$

*As in Eqn.52, the first overshoot amplitude is*

$$A_0 = \max_{k_0 \leq i \leq k_1-1} |\boldsymbol{e_v}(i)| = |\boldsymbol{e_v}(i_{max})|. \tag{53}$$

*Assume* $\sup_k \|\bar{\boldsymbol{A}}(k)\| \leq M < \infty$ *and* $\sup_k \|\boldsymbol{M}_p(k)\| \leq q < 1$. *Denote* $\|\boldsymbol{K}_i\| =: h$. *and given* $q + Mh < 1$. *Therefore,*

$$A_0 \leq Mh\Big(\frac{1}{1-q} + \frac{1}{(1-q)^2}\Big)\boldsymbol{e_v}(0) + \Big(\frac{Mh}{1-q}(k_0-1) + \frac{Mh}{1-q}\Big)\|\boldsymbol{d}\|_\infty + \frac{Mh(k_0-1)+1}{1-q}\|\boldsymbol{w}\|_\infty; \tag{54}$$

**Proof.** Before the first crossing ($j \leq k_0-1$) we have $\boldsymbol{s}_v(j) \geq 0$, hence from Eqn. 49

$$\boldsymbol{e}_v(j+1) = a(j)\boldsymbol{e}_v(j) - b(j)\boldsymbol{s}_v(j) + \boldsymbol{w}_v^{\perp}(j) \leq a(j)\boldsymbol{e}_v(j) + \boldsymbol{w}_v^{\perp}(j) \leq q\boldsymbol{e}_v(j) + \|\boldsymbol{w}\|_\infty,$$

so by induction $\boldsymbol{e}_v(j) \leq q^j \boldsymbol{e}_v(0) + \frac{1}{1-q}\|\boldsymbol{w}\|_\infty$. Summing Eqn. 50,

$$\boldsymbol{s}_v(k_0-1) = \sum_{i=0}^{k_0-2} \boldsymbol{e}_v(i) - \sum_{i=0}^{k_0-2} \boldsymbol{d}_v(i) \leq \frac{\boldsymbol{e}_v(0)}{1-q} + (k_0-1)\|\boldsymbol{d}\|_\infty + \frac{k_0-1}{1-q}\|\boldsymbol{w}\|_\infty. \tag{55}$$

Since $a(k) \geq 0$ and $\boldsymbol{e}_v(k_0-1) \geq 0$, at the crossing step,

$$\begin{aligned} |\boldsymbol{e}_v(k_0)| = -\boldsymbol{e}_v(k_0) &\leq b(k_0-1)\boldsymbol{s}_v(k_0-1) + \|\boldsymbol{w}\|_\infty \\ &\leq Mh\left(\frac{\boldsymbol{e}_v(0)}{1-q} + (k_0-1)\|\boldsymbol{d}\|_\infty + \frac{k_0-1}{1-q}W_\infty\right) + \|\boldsymbol{w}\|_\infty. \\ &= \frac{Mh}{1-q}\boldsymbol{e}_v(0) + Mh(k_0-1)\|\boldsymbol{d}\|_\infty + (Mh\frac{k_0-1}{1-q}+1)\|\boldsymbol{w}\|_\infty \end{aligned} \tag{56}$$

Assume that $\boldsymbol{d}_v(k)$ is small enough s.t during the overshoot time, $\boldsymbol{s}_v$ is nonincreasing (since $\boldsymbol{e}_v < 0$ a.e. on $[k_0, k_1-1]$), so $\boldsymbol{s}_v(i) \leq \boldsymbol{s}_v(k_0-1)$ for $i \in [k_0, k_1-1]$. Using Eqn. 49 again and unrolling $m$ steps from $k_0$,

$$\begin{aligned} |\boldsymbol{e}_v(k_0+m)| &\leq q^m |\boldsymbol{e}_v(k_0)| + \sum_{k=0}^{m-1} q^k \big(Mh\boldsymbol{s}_v(k_0-1) + \|\boldsymbol{w}\|_\infty\big) \\ &\leq |\boldsymbol{e}_v(k_0)| + \frac{Mh\boldsymbol{s}_v(k_0-1) + \|\boldsymbol{w}\|_\infty}{1-q}. \end{aligned} \tag{57}$$

Taking the maximum over $m \in \{0,1,\ldots,k_1-k_0\}$ and substituting Ineq. 55 and Ineq. 56 into Ineq. 57 yields

$$A_0 \le Mh\Big(\frac{1}{1-q}+\frac{1}{(1-q)^2}\Big)\boldsymbol{e}_v(0) + \Big(\frac{Mh}{1-q}(k_0-1)+\frac{Mh}{1-q}\Big)\|\boldsymbol{d}\|_\infty + \frac{Mh(k_0-1)+1}{1-q}\|\boldsymbol{w}\|_\infty \tag{58}$$

□

Consequently, the right-hand side of Ineq.58 is *monotone increasing in* $h$ (via the factor $Mh$) and increases as $q$ decreases (through the factors $\frac{1}{1-q}$). In particular, more aggressive PI leads to a larger first–overshoot amplitude.

**Remark 3 ("Fast-PI" specialization.)** *With the tuning used in our analysis in remark 2,* $h=\frac{(1-q)^2}{4M}$ *(the value that minimizes the comparison-system rate), so* $Mh=\frac{(1-q)^2}{4}$*. Plugging into equation 58 gives*

$$\boldsymbol{A}_0 \le \Big(\frac{1-q}{4}+\frac{1}{4}\Big)\boldsymbol{e}_v(0) + \Big(\frac{1-q}{4}(k_0-1)+\frac{1}{4}\Big)\|\boldsymbol{d}\|_\infty + \frac{k_0}{1-q}\|\boldsymbol{w}\|_\infty.$$

*In particular, in the disturbance-free case (*$\|\boldsymbol{w}\|_\infty=\|\boldsymbol{d}\|_\infty=0$*) we obtain*

$$\boldsymbol{A}_0 \le \Big(\frac{1-q}{4}+\frac{1}{4}\Big)\boldsymbol{e}_v(0),$$

*so stronger proportional action (smaller* $q$*) comes with a larger first-overshoot envelope, even though the closed-loop settles faster.*

## B.7 PID Control

### B.7.1 Stability of PID closed loop

We consider the PID update

$$\boldsymbol{u}(k) = \boldsymbol{K}_p\bar{\boldsymbol{e}}(k) + \boldsymbol{K}_i\boldsymbol{s}(k) + \boldsymbol{K}_d\big(\bar{\boldsymbol{e}}(k)-\bar{\boldsymbol{e}}(k-1)\big),$$

and define the auxiliary matrices

$$\boldsymbol{M}_p(k) := \bar{\boldsymbol{A}}(k)\big(I-\boldsymbol{K}_p\big), \qquad \boldsymbol{G}(k) := \bar{\boldsymbol{A}}(k)\boldsymbol{K}_i, \qquad \boldsymbol{H}(k) := \bar{\boldsymbol{A}}(k)\boldsymbol{K}_d,$$

together with the error increment

$$\Delta\bar{\boldsymbol{e}}(k) := \bar{\boldsymbol{e}}(k)-\bar{\boldsymbol{e}}(k-1), \qquad \Delta\bar{\boldsymbol{e}}(-1)=0$$

Using the plant relation, we obtain

$$\Delta\bar{\boldsymbol{e}}(k+1) = \big(\boldsymbol{M}_p(k)-I\big)\bar{\boldsymbol{e}}(k) - \boldsymbol{G}(k)\tilde{\boldsymbol{s}}(k) - \boldsymbol{H}(k)\Delta\bar{\boldsymbol{e}}(k) + \boldsymbol{w}^{\perp}(k), \tag{59}$$

Introduce the lifted state from the auxiliary PI state in Eqn. 41

$$\tilde{\zeta}_{\mathrm{PID}}(k) := \begin{bmatrix} \bar{\boldsymbol{e}}(k) \\ \tilde{\boldsymbol{s}}(k) \\ \Delta\bar{\boldsymbol{e}}(k) \end{bmatrix},$$

Then the closed-loop evolution reads

$$\tilde{\zeta}_{\mathrm{PID}}(k+1) = \boldsymbol{M}_d(k)\tilde{\zeta}_{\mathrm{PID}}(k) + \tilde{\boldsymbol{w}}_{\mathrm{PID}}(k), \tag{60}$$

where

$$\boldsymbol{M}_d(k) := \begin{bmatrix} \boldsymbol{M}_p(k) & -\boldsymbol{G}(k) & -\boldsymbol{H}(k) \\ I & I & 0 \\ \boldsymbol{M}_p(k)-I & -\boldsymbol{G}(k) & -\boldsymbol{H}(k) \end{bmatrix}, \qquad \tilde{\boldsymbol{w}}_{\mathrm{PID}}(k) := \begin{bmatrix} \boldsymbol{w}^{\perp}(k) \\ -d(k) \\ \boldsymbol{w}^{\perp}(k) \end{bmatrix}.$$

**Theorem 1 (Stabilizing the PID loop preserves bias removal)** *Let* $\boldsymbol{M}_p(k) = \bar{\boldsymbol{A}}(k)(\boldsymbol{I}-\boldsymbol{K}_p)$*, and denote* $\|\boldsymbol{K}_i\| =: h$*,* $\|\boldsymbol{K}_d\| =: \ell$*. Assume* $\sup_k\|\bar{\boldsymbol{A}}(k)\| \le M < \infty$ *and* $\sup_k\|\boldsymbol{M}_p(k)\| \le q < 1$*. If* $q+Mh<1$ *(stable PI loop), then there exists* $\ell>0$ *such that the PID closed-loop control is ISS. Therefore, the integral part in PID design still cancels the matched disturbance component* $\boldsymbol{w}^{\parallel}$*.*

**Proof.** We establish the ISS for system 60 using the method of ISS-Lyapunov function, see (Jiang et al., 1999, Def. 2.2, Prop. 2.3). It then suffices to construct a candidate ISS-Lyapunov function $\boldsymbol{V}_{\mathrm{PID}}(k)$ satisfying that there exist class $\mathcal{K}_\infty$ functions $\alpha_1,\alpha_2,\alpha_3$ and a class $\mathcal{K}$ function $\sigma$ such that

$$\alpha_1(\|\tilde{\zeta}_{\mathrm{PID}}(k)\|) \le \boldsymbol{V}_{\mathrm{PID}}(\tilde{\zeta}_{\mathrm{PID}}(k)) \le \alpha_2(\|\tilde{\zeta}_{\mathrm{PID}}(k)\|), \tag{61}$$

and

$$\boldsymbol{V}(\tilde{\zeta}_{\mathrm{PID}}(k+1)) - \boldsymbol{V}(\tilde{\zeta}_{\mathrm{PID}}(k)) \le -\alpha_3(\|\tilde{\zeta}_{\mathrm{PID}}(k)\|) + \sigma(\|\boldsymbol{w}\|). \tag{62}$$

Step 1: Candidtate $\boldsymbol{V}_{PID}(k)$ and PI-closed loop baseline

Define

$$\boldsymbol{V}_{\mathrm{PID}}(k) := \boldsymbol{V}_{\mathrm{PI}}\big(\tilde{\zeta}_{\mathrm{PI}}(k),k\big) + r\|\Delta\bar{\boldsymbol{e}}(k)\|^2, \qquad r>0, \tag{63}$$

where $\boldsymbol{V}_{\mathrm{PI}}(\tilde{\zeta}_{\mathrm{PI}},k) = \tilde{\zeta}_{\mathrm{PI}}^\top \boldsymbol{P}(k)\tilde{\zeta}_{\mathrm{PI}}$ with $\boldsymbol{P}(k) = \boldsymbol{P}(k)^\top \succ 0$ and there exists some $\mu_{\mathrm{PI}} > 0$ such that

$$\boldsymbol{M}_i(k)^\top \boldsymbol{P}(k)\boldsymbol{M}_i(k) - \boldsymbol{P}(k) \le -\mu_{\mathrm{PI}} I, \qquad \forall k, \tag{64}$$

Regarding the existence of such $\boldsymbol{V}_{\mathrm{PI}}$, recall the homogeneous PI-loop $\tilde{\zeta}_{\mathrm{PI}}(k+1) = \boldsymbol{M}_i(k)\tilde{\zeta}_{\mathrm{PI}}(k)$ with $\boldsymbol{M}_i(k) = \begin{bmatrix} \boldsymbol{M}_p(k) & -\boldsymbol{G}(k) \\ I & I \end{bmatrix}$, being asymptotically stable. Suppose there is $\boldsymbol{Q}(k) = \boldsymbol{Q}(k)^\top \succeq 0$ bounded so that the pair $(\boldsymbol{M}_i(k), \sqrt{\boldsymbol{Q}(k)})$ is observable for all $k$, hence the difference Lyapunov equation

$$\boldsymbol{M}_i^\top(k)\boldsymbol{P}(k+1)\boldsymbol{M}_i(k) - \boldsymbol{P}(k) = -\boldsymbol{Q}(k)$$

admits a unique positive definite solution $\boldsymbol{P}(k) = \boldsymbol{P}^\top(k) \succ 0$ for all $k$, and a uniform bound $\|\boldsymbol{P}\|_\infty := \sup_k \|\boldsymbol{P}(k)\| < \infty$ (see (Gajic & Qureshi, 2008, Ch. 4, p. 110))

Step 2: Condition (i) as in Eqn. 61

We write $\boldsymbol{V}_{\mathrm{PID}}$ as a quadratic form

$$\boldsymbol{V}_{\mathrm{PID}}(k) = \tilde{\zeta}_{\mathrm{PID}}(k)^\top \boldsymbol{P}_*(k)\tilde{\zeta}_{\mathrm{PID}}(k), \qquad \boldsymbol{P}_*(k) := \begin{bmatrix} \boldsymbol{P}(k) & 0 \\ 0 & rI \end{bmatrix}. \tag{65}$$

Clearly $\boldsymbol{P}_*(k) = \boldsymbol{P}_*(k)^\top$ and $\boldsymbol{P}_*(k) \succ 0$ because $\boldsymbol{P}(k) \succ 0$ and $rI \succ 0$; hence $\boldsymbol{P}_*(k)$ is symmetric positive definite for all $k$.

By the spectral theorem, there exists an orthogonal matrix $U_*(k)$ and a diagonal $\Lambda_*(k) = \mathrm{diag}(\lambda_1(k), \dots, \lambda_{n_*}(k))$ with positive entries such that $\boldsymbol{P}_*(k) = U_*(k)\Lambda_*(k)U_*(k)^\top$. Moreover, because $\boldsymbol{P}_*(k)$ is block diagonal, its eigenvalues are precisely the union of the eigenvalues of $\boldsymbol{P}(k)$ and the repeated eigenvalue $r$. Using the uniform bounds already established for $\boldsymbol{P}(k)$ (there exists $\underline{\lambda} > 0$ with $\lambda_{\min}(\boldsymbol{P}(k)) \ge \underline{\lambda}$ and $\lambda_{\max}(\boldsymbol{P}(k)) \le \|\boldsymbol{P}\|_\infty := \sup_k \|\boldsymbol{P}(k)\| < \infty$), we obtain the $k$-independent bounds

$$\lambda_{\min}\big(\boldsymbol{P}_*(k)\big) \ge \underline{\lambda}_* := \min\{\underline{\lambda}, r\} > 0, \qquad \lambda_{\max}\big(\boldsymbol{P}_*(k)\big) \le \overline{\lambda}_* := \max\{|\boldsymbol{P}\|_\infty, r\} < \infty.$$

For every vector $z$ and every symmetric positive definite $M$, $\lambda_{\min}(M)\|z\|^2 \le z^\top M z \le \lambda_{\max}(M)\|z\|^2$. Applying this to $M = \boldsymbol{P}_*(k)$ and $z = \tilde{\zeta}_{\mathrm{PID}}(k)$ in Eqn. 65 yields

$$\underline{\lambda}_* \big\|\tilde{\zeta}_{\mathrm{PID}}(k)\big\|^2 \le \boldsymbol{V}_{\mathrm{PID}}(k) \le \overline{\lambda}_* \big\|\tilde{\zeta}_{\mathrm{PID}}(k)\big\|^2.$$

Therefore, choosing the class-$\mathcal{K}_\infty$ functions

$$\alpha_1(\boldsymbol{s}) := \underline{\lambda}_* \boldsymbol{s}^2, \qquad \alpha_2(\boldsymbol{s}) := \overline{\lambda}_* \boldsymbol{s}^2,$$

we obtain the desired bound

$$\alpha_1\Big(\big\|\tilde{\zeta}_{\mathrm{PID}}(k)\big\|\Big) \le \boldsymbol{V}_{\mathrm{PID}}(k) \le \alpha_2\Big(\big\|\tilde{\zeta}_{\mathrm{PID}}(k)\big\|\Big), \tag{66}$$

which establishes condition (i) in 61.

Step 3: Condition (ii) as in Eqn. 62.

From Eqn. 63,

$$\Delta\boldsymbol{V}_{\mathrm{PID}}(k) = \underbrace{\Delta\boldsymbol{V}_{\mathrm{PI}}\Big(\tilde{\zeta}_{\mathrm{PI}}(k)\Big)\Big|_{\mathrm{PID}}}_{\text{PI part under PID update}} + r\big(\|\Delta\bar{\boldsymbol{e}}(k+1)\|^2 - \|\Delta\bar{\boldsymbol{e}}(k)\|^2\big). \tag{67}$$